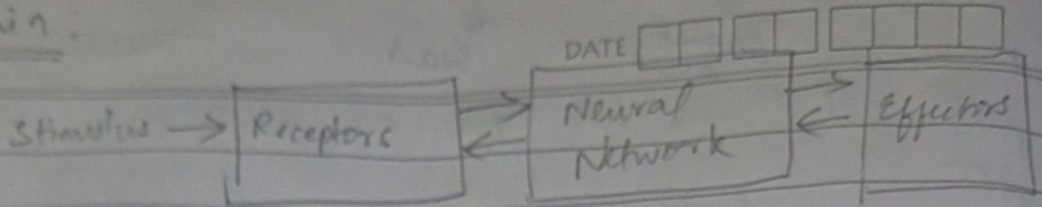


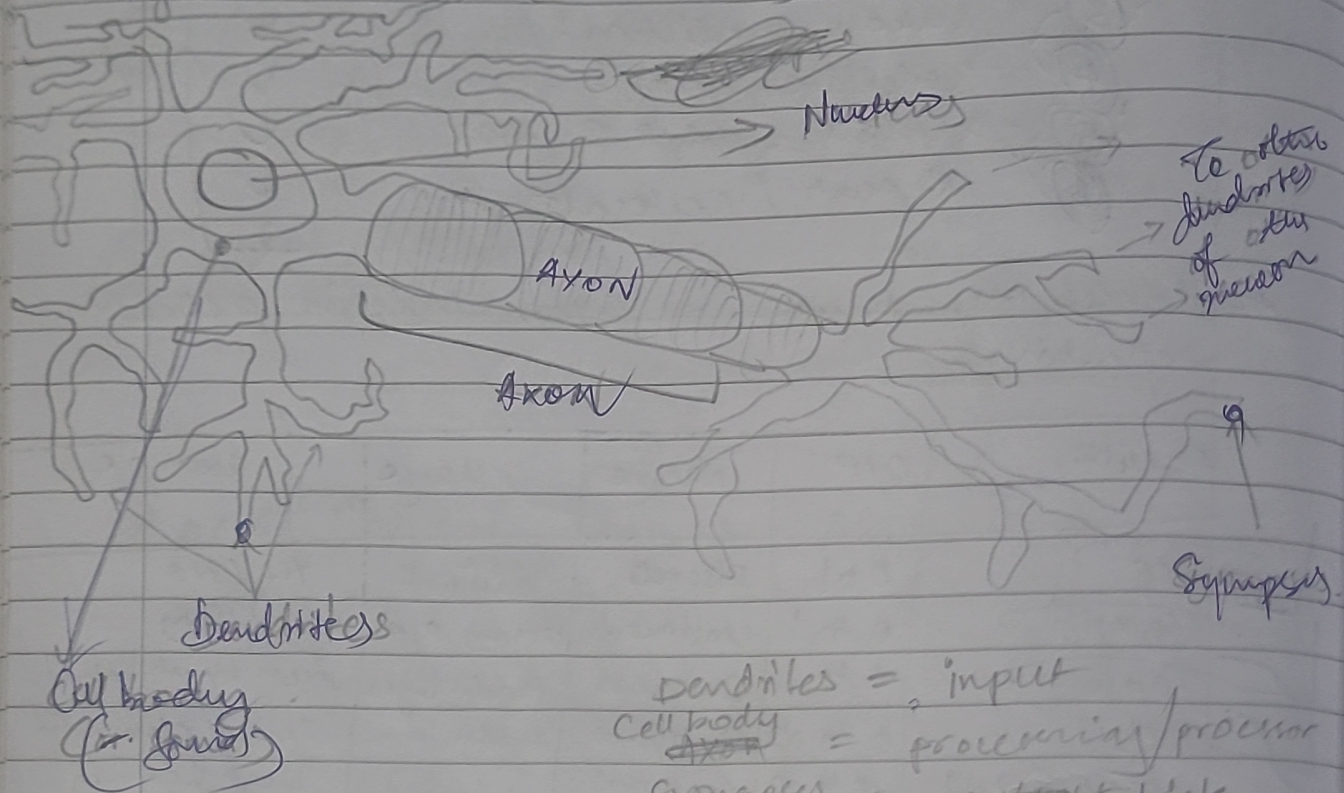
Human Brain



Features of the Brain:

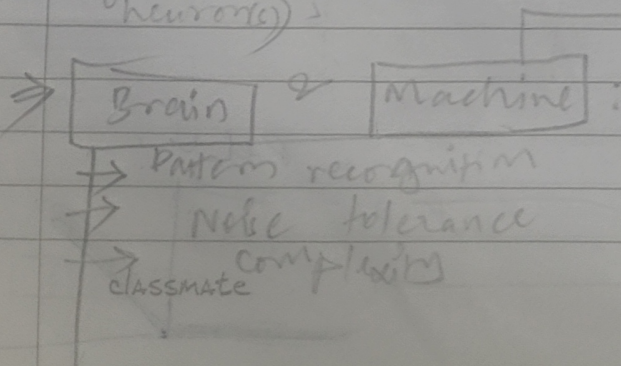
- It is massively parallel information processing system
- 10^{10} neurons (10 Billion Nerve cells)
- On average, several thousand connections.
- Hundreds of operations per second

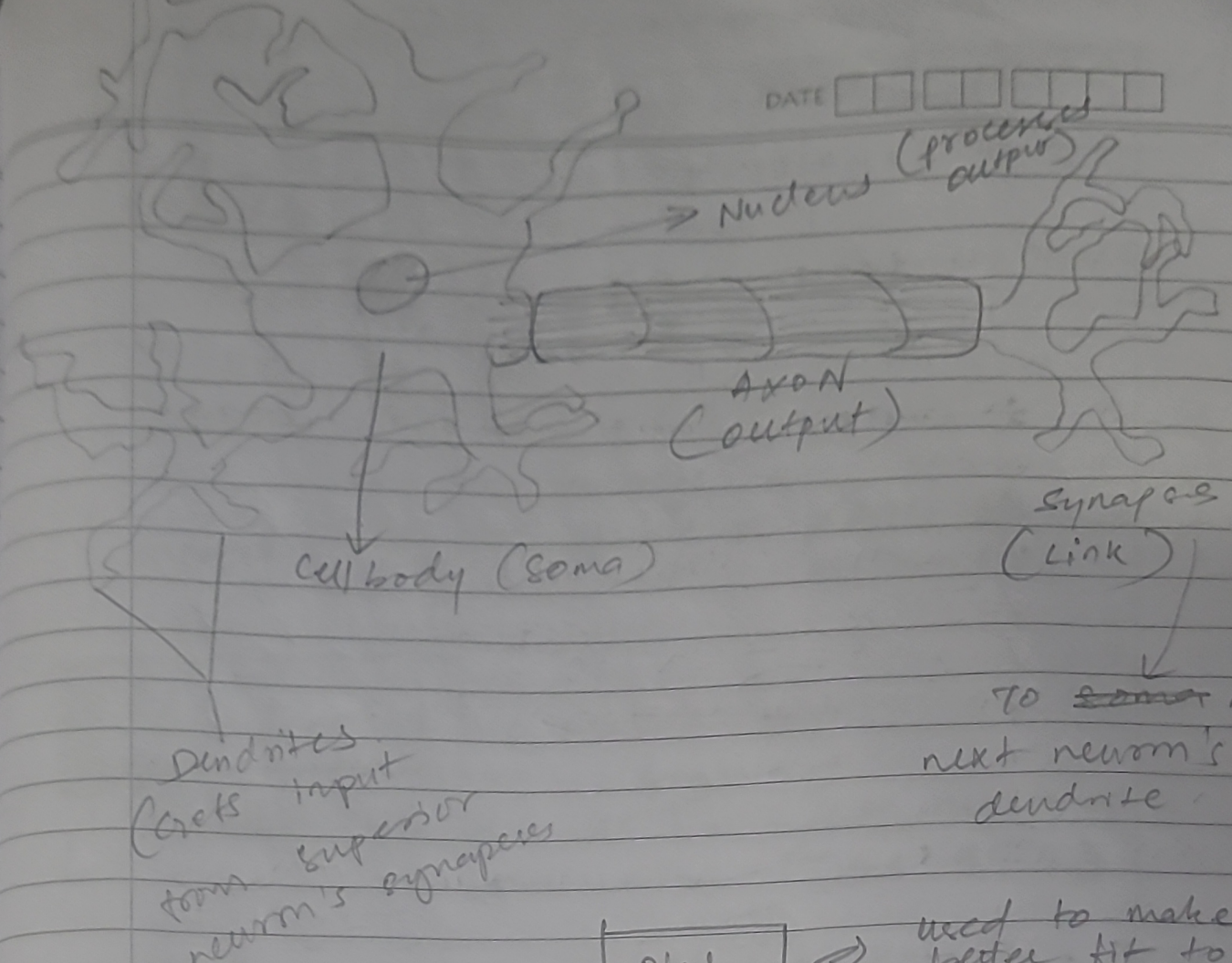
Biological Neuron (Nerve cell).



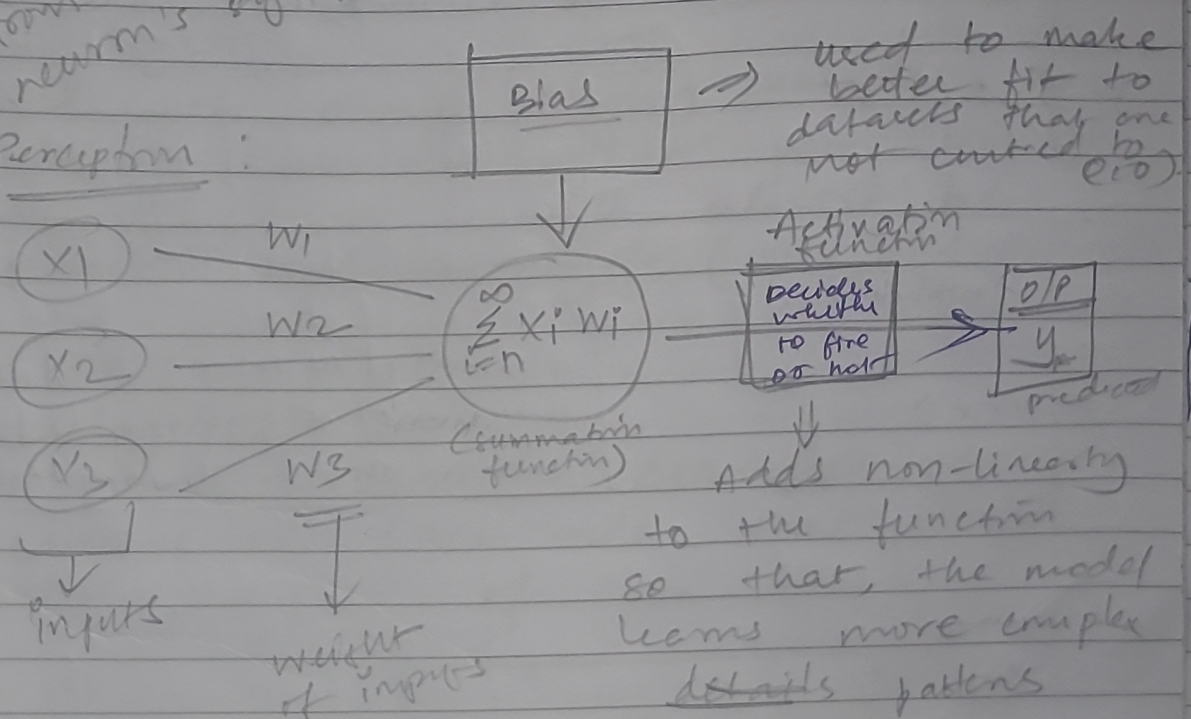
Dendrites = input
 Cell body = processing/processor
 Synapses = ~~input~~ link
 Axon = Output

- once input to neuron exceeds a critical level, the neuron discharges a spike (an electrical impulse that travels from the body, down to the rest of the neuron).





→ Perception :



~~Without Bias~~ ^{Bias}, the ~~decision~~ helps to push the threshold. It helps the program to make better decisions.

$$y = \sum_{i=1}^n x_i w_i + B$$

Threshold = 1.5

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$$y = \left(\sum x_i w_i + b \right)$$

$$(0.9)(1) + 0 + 0.5 + 0 + 0.4 + b$$

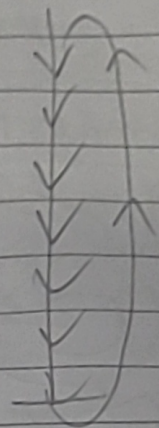
$$= 1.6 \geq 1.5 \quad \checkmark$$

input $\rightarrow$ $\sum x_i w_i$ $\rightarrow$ Summation $\rightarrow$ bias $\rightarrow$ output
 activation function

The activation function outputs output is always b/w 0 to 1 Yes

$\Rightarrow$ Training a perceptron :- or adjust

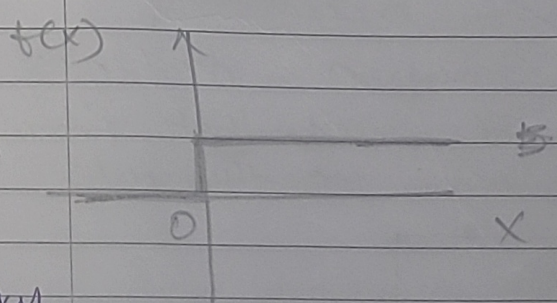
- ① Initialize weights & bias
- ② Input the training data
- ③ Make prediction
- ④ Calculate Error
- ⑤ Update weights & bias
- ⑥ Repeat / Reiterate



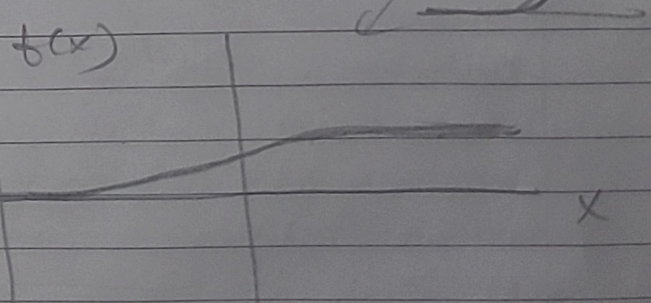
* Perceptron uses a step function activation function.

Function	O/P	Nature	usage	example
Step	0 to 1	Discrete	Simple classification	"Yes / No"
Sigmoid	0 to 1	Continuous	Binary o/p	"Probability"
Tanh	-1 to 1	Continuous	Hidden layer	"Keeps signals balanced"
ReLU	0 to ∞	Continuous	Hidden layer	"Image classification"
Leaky ReLU	$-\infty$ to ∞	Continuous	" "	"Deep Learning"

Step function:

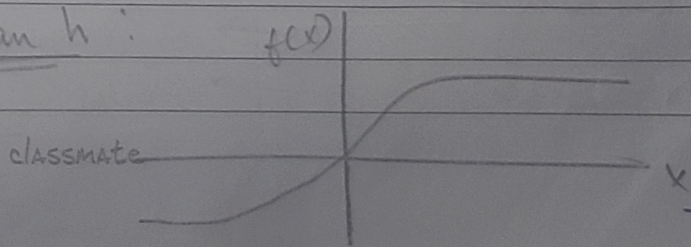


Sigmoid function:

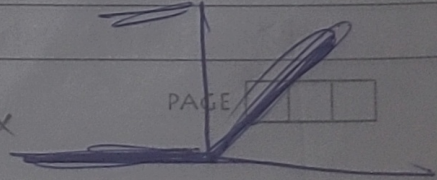


Leaky ReLU $\rightarrow$

Tanh:



ReLU:



Jan 11

Back propagation

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• learn from mistakes & improve at making accurate predictions

Types of Neural Network :-

① Feedforward NN.

Signals move only forward

Input $\rightarrow$ output

eg: Image recognition, basic classification

② CNN: (Convolutional neural network)

Looks at patches / small sections

eg: Facial recognition, object detection

③ RNN (Recurrent N.N.) Loops

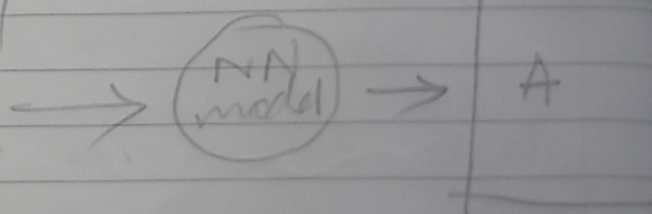
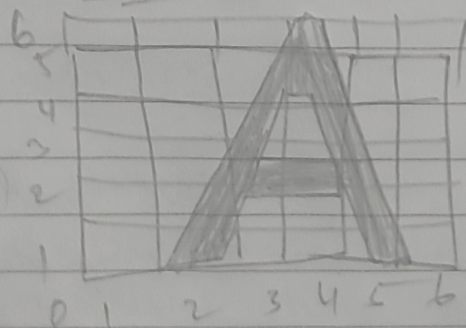
It loops, remembers past data

eg: Text prediction, language translation

$\Rightarrow$ OCR (Optical Character Recognition)

A technology that converts a image into machine-readable text using neural network models.

6x6 OCR:



→ Neural Network :-

Mimics human biological system
to solve complex problems.

- * Parallel Distributed processing
- * Neural Computation

structure of neurons: fires when the

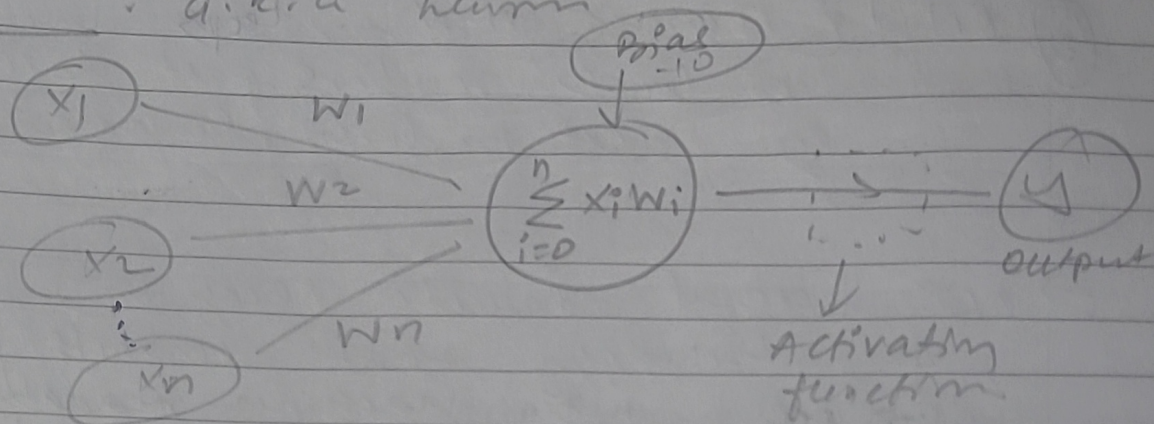
* A neuron ~~only~~ fires if input signal exceeds a certain threshold in a short time period.

* Synapses vary in length

→ Good connection = strong signal

→ Loose connection = weak signal.

Perceptron: a.k.a. neuron



→ presented to the neuron

Inputs = ~~x1~~ $x_1, x_2, \dots, x_n$

Weights = $w_1, w_2, \dots, w_i$

Bias = -10

Output = y .

Bias: * used to ~~fit~~ do better fit

for a model to the datasets

* Improves model flexibility

* It ensures model can fit

dataset that are not centred towards (0,0),

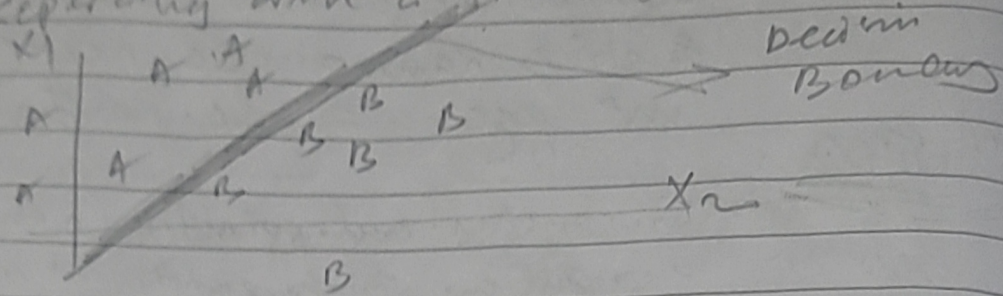
Activating function

classmate

adds a non-linearity to the model, to learn non complex patterns.

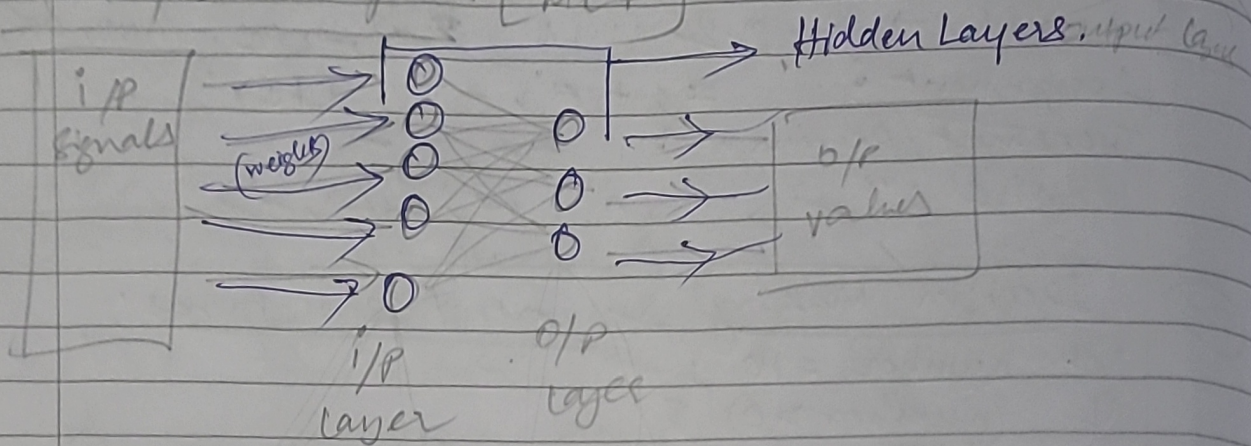
Under separability

separating with a linear line



Training Algorithm:

Multilayer Perceptron [MLP]



Hidden layer:

- performs classification of features
- Two hidden layers are sufficient
- Features imply more layers may be better

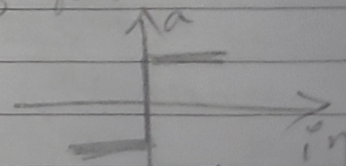
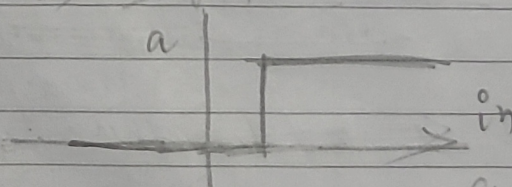
→ Activation Functions!

* Transforms neuron's i/p to o/p

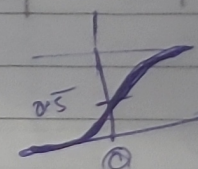
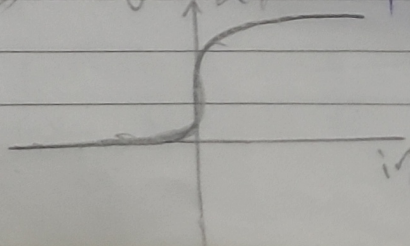
types

a) Step function

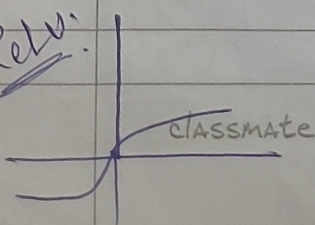
b) sign function



c) Sigmoid function



Relu:



Linear: the o/p is proportional to the total weight

Transfer functions:

- Linear - The o/p is o/p & weighted i/p.
- Non-linear - o/p varies continuously, they are not linear as i/p changes.
- Threshold - Amount / quantity which is exceeded, will activate (fire) the neurons.

→ Backpropagation :-

- * an example of supervised learning technique.
- * used to minimize the error.

N = neuron.

N_W = Neuron's i/p weight

$N_{out} =$ (1) ~~o/p~~ Neuron's output

$$N_W = N_W + \Delta N_W$$

$$\Delta N_W = N_{out} * (1 - N_{out}) * N_{ErrorFactor}$$

$$\Delta N_{ErrorFactor} = N_{expected} - N_{Actual}$$

→ Learning Types :-

① Hebbian Learning : (Piano learning)

~~describes~~ * This learning reflects brain plasticity, like "How the brain improves when doing a task again & again like playing a piano."

* When a task is repeated, the neurons fire together frequently, which increases the synaptic weights - improving skill performance.

Competitive Learning: Fastest & best neuron is selected.

Two new

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the neuron learns each time.

the fastest & the best neuron is selected to learn.

Neuron A

Neuron A

Neuron B

W_A

$$W_A = [0.2, 0.6]$$

$$W_B = [0.8, 0.4]$$

$$\text{Input } x = [0.3, 0.5]$$

$$d_A = \sqrt{(0.3 - 0.2)^2 + (0.5 - 0.6)^2} = \sqrt{0.02} = 0.141$$

$$d_B = \sqrt{(0.3 - 0.8)^2 + (0.5 - 0.4)^2} \\ = \sqrt{0.26} \approx 0.510$$

$$d_A = 0.141 \quad d_B = 0.510$$

$$0.141 < 0.510$$

$$d_A < d_B$$

∴ Therefore Neuron A is selected.

Hebbian Learning: (Plano)

If two neurons are active at same time, then, the connection gets stronger.

x i/p with activity x

y o/p with activity y

connection weight b/w them = w

Hebbian learning rule:

$$\Delta W = \eta \cdot x \cdot y$$

η = learning rate,
(small +ve number)

→ Booth's algorithm :
 * used for multiplication of signed
 binary numbers.

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→ Euclidean Algorithm :
 Find HCF & LCM of two integers

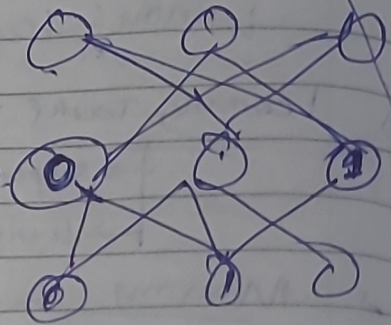
→ Boltzman Learning :

* use probabilistic approach to model the data.

* They are energy based models

* Each neuron could be in binary state

Ex. Image recognition
 NLP, Recommendation systems



→ Formulas :

① ~~we have~~ $W_{new} = W_{old} + \eta (d - y) x$

$W_{new} = W_{old} + \eta (d - y) x$
 η : learning rate
 d : desired output
 y : actual output

② Hebbian Learning Rule :

$$\Delta W = \eta \cdot x \cdot y$$

~~$\Delta W = 0$~~

③ Competitive learning :

$$W_{new} = W_{old} + \eta (x - W_{old})$$

④ Boltzman Learning :

$$\Delta W = \eta (p^+ - p^-)$$

⑤ Sigmoid Neuron output :

$$\text{sigmoid } f(z) = \frac{1}{1 + e^{-z}}$$

$$z = \sum w_i x_i + b$$

PAGE

Credit Assignment problem :-

- * When an agent performs a series of actions, its supposed to given rewards (credits) DATE
- * These credits are given only to the actions that led the agent to the goal.
- * Unnecessary steps or the useless steps are given penalties not the credits such that the agent will learn
- * The agent must figure out which actions in the past led to the goal & which did not lead at all.

Learning with/without teacher:-

Type	Teacher (labelled data)	
With teacher	Yes	Supervised
Without teacher	NO	Unsupervised

Learning Tasks:

- supervised (Classification, Regression)
- unsupervised (Clustering, Association)

* Memory in NN:

- * Ability of the NN to retain/recall the past data or experience

→ Adaptability:

- Ability of the model to dynamically adapt & or predict unseen data